HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2025)

Student No.: 21 Name: _____ Marks: / 66

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Simplify $\frac{(x^4y^{-5})^3}{xy^2}$ and express your answer with positive indices. (3 marks) (DSE-2025 PI Q01)

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2. The selling price of a souvenir is $378. The souvenir is sold at a discount of 60% on its marked price. The marked price of the souvenir is 75% above its cost.

(a) Find the marked price of the souvenir.

(b) Find the cost of the souvenir.

(c) Determine whether there is a gain or a loss after selling the souvenir. Explain your answer.

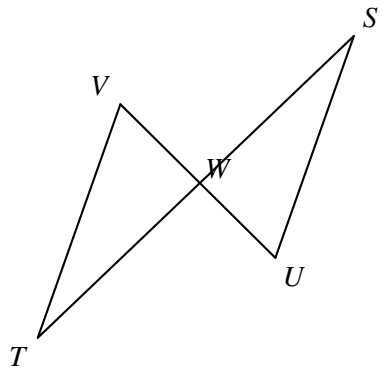
(5 marks) (DSE-2025 PI Q07)

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3. In the figure, ST and UV intersect at the point W. It is given that $SU \parallel VT$ and W is the mid-point of ST.



(a) Prove that $\triangle SUW \cong \triangle TVW$.

(b) Let X be a point lying on TW such that $\triangle SUW \sim \triangle VXW$. If $SU = 57\,\text{cm}$, $SW = 63\,\text{cm}$ and $WX = 7\,\text{cm}$, find the perimeter of $\triangle TVX$.

(5 marks) (DSE-2025 PI Q08)

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4. The table below shows the distribution of the numbers of pens owned by some students.

Number of pens	5	6	7	8	9
Number of students	9	13	2	s	11

The median of the distribution is 7.

(a) Find the least possible value of s.

(b) Write down the greatest possible value of s.

(c) Find the greatest possible standard deviation of the distribution.

(5 marks) (DSE-2025 PI Q09)

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5. Let $f(x) = 2x^3 + hx^2 + kx + 15$, where h and k are constants. When $f(x)$ is divided by $x+2$, the remainder is -45. It is given that $f(x)$ is divisible by $2x-5$.

(a) Find h and k.

(b) Someone claims that all the roots of the equation $f(x) = 0$ are rational numbers. Do you agree? Explain your answer.

(6 marks) (DSE-2025 PI Q10)

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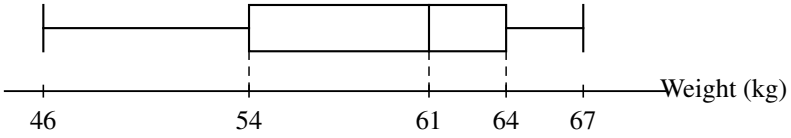
6. The stem-and-leaf diagram below shows the distribution of the weights (in kg) of some athletes before training.

Stem (tens)	Leaf (units)
4	5 6 8
5	w w 6 7
6	3 4 5 6 9 9
7	0 2
8	7

The difference of the range and the inter-quartile range of the above distribution is 25 kg.

(a) Write down the range of the above distribution. Hence, find w.

(b) The box-and-whisker diagram below shows the distribution of the weights (in kg) of the athletes after the training.



(i) Find the change in the upper quartile of the distribution due to the training.

(ii) Is the distribution of the weights of the athletes after the training less dispersed than that before the training? Explain your answer.

(7 marks) (DSE-2025 PI Q12)

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7. There are 8 white plates, 4 red plates and 1 black plate in a box. If 3 plates are randomly drawn from the box at the same time, find

(a) the probability that the colours of the 3 plates drawn are the same;

(b) the probability that the colours of the 3 plates drawn are all different.

(4 marks) (DSE-2025 PI Q15)

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8. It is given that $\begin{cases} \log_3 x + \log_3 y = 9 \\ \log_x 81 - \log_y 9 = 1 \end{cases}$, where $0 < x < y$.

(a) Let $u = \log_3 y$. Prove that $u^2 - 3u - 18 = 0$.

(b) Find x.

(4 marks) (DSE-2025 PI Q16)

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9. Let $T(n)$ be the nth term of an arithmetic sequence such that $T(1) \neq T(2)$ and $T(47) = 456$. It is given that $T(9)$, $T(47)$, $T(199)$ is a geometric sequence.

(a) Find $T(1)$.

(b) Find the least value of n such that the sum of the first n terms of the arithmetic sequence is greater than 10^6.

(6 marks) (DSE-2025 PI Q17)

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10. Let $g(x) = 3x^2 - 6kx + 24x + 3k^2 - 24k + 55$, where k is a positive constant. Denote the vertex of the graph of $y = g(x)$ by R.

(a) Using the method of completing the square, express the coordinates of R in terms of k.

(b) The graph of $y = h(x)$ is obtained by translating the graph of $y = g(x)$ rightwards by 6 units and then downwards by 10 units. Denote the vertex of the graph of $y = h(x)$ by S. Let T be a point such that the orthocentre of $\triangle RST$ is the origin.

(i) Express the coordinates of T in terms of k.

(ii) The coordinates of the point U are $(-5, 5)$. Let V be a point lying on RS such that TV is perpendicular to RS. It is given that S, T, U and V are concyclic. Find k.

(9 marks) (DSE-2025 PI Q18)

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11. The equation of the straight line L is $4x-3y+83=0$. The equation of the circle C is $x^2+y^2+ax-2y+b=0$, where a and b are positive constants. It is given that C passes through the point $(-10,9)$ and L is a tangent to C.

(a) Prove that the equation $25x^2+(9a+640)x+(90a+4924)=0$ has equal roots. Hence, find a and b.

(b) The coordinates of the point P are $(-20,1)$. Let Q be a point lying below L such that PQ is tangent to C and $PQ=25$. Denote the centre of C by I.

(i) Find $\cos\angle IPQ$.

(ii) Find, in surd form, the distance between I and Q.

(iii) Let R be a point lying on L such that C is the inscribed circle of $\triangle PQR$. Denote the radius of the circumcircle of $\triangle PQR$ by r. Someone claims that $r>PQ$. Is the claim correct? Explain your answer.

(12 marks) (DSE-2025 PI Q19)

I can finish _____ questions

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